

Phase D2 Key Prior-Art Claims Index

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Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

This note is a navigation reference for the Phase D2 prior-art corpus. Phase D2 comprises notes D2.01 through D2.78, establishing operational variants and decompositions of the Paper 3 inter-Self coordination architecture. This index organizes the most important prior-art claims those notes collectively establish into fourteen named claims across four categories: Architectural (AC-1 through AC-5), Operational (OC-1 through OC-5), Governance Quality (GC-1 through GC-5), and Population-Scope (PC-1 through PC-3). Each claim entry states what the claim establishes, its prior-art significance, and the specific D-series notes that establish it. The index is designed for use as a reference tool: a reader locating a claim for litigation support, prior-art search, or subsequent derivation work can find the establishing notes directly from this document. Note D2.80 follows as the final Phase D2 note, providing full synthesis and formal closure of the phase.

1. Purpose and Scope of This Index

Phase D2 is a corpus of seventy-eight operational derivation notes (D2.01–D2.78) that decompose the Paper 3 inter-Self coordination architecture into specific, individually defensible prior-art claims. The corpus spans governance sequences, compliance tests, anti-patterns, boundary cases, and operational properties across the full scope of the Full Aspect Integration (FAI) architecture. No single note within the corpus provides a consolidated view of what, collectively, Phase D2 establishes.

This index supplies that consolidated view. Its primary function is reference, not argument. The establishing note references — the specific D-series note IDs cited under each claim — are the

document's primary value. Each cited note provides the formal derivation, operational detail, and prose argument that establishes the claim; this index names the claim, states it concisely, and points to where the full establishment lives.

The fourteen claims indexed here are affirmative claims — what the architecture commits to, what governance must do, what properties the architecture guarantees, and what calibrated-humility positions the corpus takes at population scope. The adversarial closure work that anticipates and rebuts challenges to these claims was handled separately in D2.77. An adversary challenging the Phase D2 prior-art corpus must address all fourteen affirmative claims stated here, each traceable to specific establishing notes.

The four categories reflect the structure of what Phase D2 covers:

Architectural claims (AC-1–AC-5): What the inter-Self coordination architecture commits to by construction — inherited commitments from Papers 1 and 2 that hold at inter-Self scope.

Operational claims (OC-1–OC-5): What governance must do to operate a compliant FAI event — specific enumerations of requirements, records, tests, and failure modes.

Governance quality claims (GC-1–GC-5): What properties the architecture guarantees about the quality and character of governance — properties that hold by design, not by procedural convention.

Population-scope claims (PC-1–PC-3): What the architecture commits to at population scale, stated in the calibrated-humility register appropriate for claims that are architectural in character but population-empirical in verification.

2. Category 1 — Architectural Prior-Art Claims (AC-1 through AC-5)

Architectural claims state what the inter-Self coordination architecture commits to by construction. They are the structural prior-art foundation on which the operational, governance quality, and population-scope claims rest. Each architectural claim is established by deriving it from the Paper 1 and Paper 2 commitments it inherits, then demonstrating that the inheritance holds at inter-Self scope.

AC-1: Shared Substrate as Paper 1 Substrate at Inter-Self Scope

What the claim establishes. The shared substrate that hosts FAI is a Paper 1 substrate operating at inter-Self scope. All six Paper 1 architectural commitments — substrate as coordination artifact, conflict preservation as first-class property, human-governed authority, AI as substrate mediator, tool-agnosticism, and linear-cost composition — hold within the shared substrate. Paper 3 does not introduce a new substrate concept; it extends the Paper 1 substrate commitment to a larger scope while preserving all six commitments without modification.

Prior-art significance. This claim blocks any attempt to patent the shared substrate of FAI as a novel substrate architecture. The shared substrate is not novel — it is a Paper 1 substrate instantiated at a new scope. Any claim of novelty for the shared substrate must contend with the Paper 1 prior art and with this explicit scope-extension derivation.

Establishing notes: D1.02, D2.03, D2.04, D2.05, D2.76

AC-2: FAI as Paper 2 Mating Primitive at Inter-Self Scope

What the claim establishes. Full Aspect Integration is the Paper 2 mating primitive operating at inter-Self scope. FAI exhibits the same three pattern variants (full merge, selective merge, conflict-preserving exchange) that Paper 2's mating primitive exhibits within a single Self. FAI preserves conflicts as first-class substrate content, following the same conflict-preservation commitment Paper 2 inherits from Paper 1. The difference between mating and FAI is scope and outcome: mating produces offspring or evolved forms within a Self; FAI produces evolution outputs that re-enter the participating Selves' home evolution machinery.

Prior-art significance. This claim blocks any attempt to patent FAI as a novel coordination or merge primitive. FAI is the Paper 2 mating primitive at a different scope. Any novelty claim for FAI as a primitive must contend with the Paper 2 mating prior art and with this scope-extension derivation.

Establishing notes: D1.12, D2.07, D2.08, D2.09

AC-3: Three-Tier Conflict Handling Inherited from Papers 1 and 2

What the claim establishes. The three-tier conflict-handling architecture at inter-Self scope — preserve at substrate level, resolve via orchestration with inspectable rules, escalate to humans across joint authority — inherits from Paper 1's conflict-as-first-class-property commitment and Paper 2's recursive conflict-handling derivation. No new conflict resolution mechanism is required at inter-Self scope. The three tiers are the same three commitments operating at a larger perimeter. The escalation path crosses organizational boundaries, but the architectural structure of the escalation is unchanged from the Paper 1 and Paper 2 prior art.

Prior-art significance. This claim blocks any attempt to patent inter-organizational conflict-handling in shared substrate architectures as novel. The three-tier structure is derived from prior art; its operation at inter-Self scope is an extension, not an invention.

Establishing notes: D0.03, D1.13–D1.15, D2.13, D2.14, D2.51

AC-4: Evolution Feed via Existing Paper 2 Mechanisms

What the claim establishes. The evolution feed produced by FAI uses Paper 2's existing evolution mechanisms. FAI outputs enter the participating Selves' home evolution machinery through substrate ingestion — the same per-mechanism feed structure, the same three-case layer-routing rule, and the same asymmetric ingestion commitment that Paper 2 establishes for intra-Self evolution. FAI does not add new evolution mechanisms; it adds a new locus (FAI dissolution hand-off) at which existing mechanisms receive inputs.

Prior-art significance. This claim blocks any attempt to patent the FAI-to-evolution pathway as a novel evolution mechanism. The pathway uses Paper 2 mechanisms operating at their established ingestion interfaces. The novelty in Paper 3 is the inter-Self coordination that produces evolution-eligible outputs — not the mechanisms that process those outputs once produced.

Establishing notes: D0.04, D1.17–D1.21, D2.10, D2.11, D2.52, D2.57

AC-5: FAI Configuration as Substrate Content (Claim 5)

What the claim establishes. FAI configuration — participation rules, aspect exchange scope, conflict-handling rules, dissolution conditions, and all other configurable dimensions of an FAI event — is substrate content in the Paper 1 and Paper 2 sense of that term. Configuration lives in the shared substrate and is subject to the full set of Paper 1 governance commitments: human inspection rights, modification rights, and override rights. The configuration commitment is recursive: configuration-of-configuration is also substrate content. This recursion bottoms at human-authored authority.

Prior-art significance. This claim blocks any attempt to patent configurable FAI parameters as a novel governance layer or configuration architecture. FAI configuration is not architecturally separate from the substrate — it is substrate content, and the Paper 1 Claim 5 prior art (configuration as substrate content, derived in D0.05) covers it.

Establishing notes: D0.05, D1.22–D1.25, D2.12, D2.51, D2.52

3. Category 2 — Operational Prior-Art Claims (OC-1 through OC-5)

Operational claims state what governance must do to operate a compliant FAI event. They are enumerative claims — the prior art consists not just in the existence of requirements, records, tests, and failure modes, but in the specific numbered enumerations that give the corpus its coverage density. Each operational claim names a specific count and provides a reference to the note that establishes the complete enumeration.

OC-1: Minimum Viable Governance Floor — Seven Irreducible Requirements

What the claim establishes. A minimum viable governance floor of seven irreducible requirements for a compliant FAI event, covering: (1) shared substrate present and governed, (2) participation explicitly authorized in substrate content, (3) aspect exchange scope defined and bounded, (4) conflict-handling tier assignment specified, (5) dissolution conditions specified in advance, (6) evolution feed routing specified for each output type, and (7) governance record adequate for deterministic retraceability. No compliant FAI event can omit any of the seven requirements. The floor is irreducible: removing any single requirement leaves the FAI event without architectural coverage for at least one Paper 3 commitment.

Prior-art significance. This claim establishes a specific, named minimum for FAI governance compliance. Any governance framework claiming to support FAI coordination must meet this floor or document a principled architectural reason why a given requirement does not apply. The irreducibility argument is prior art.

Establishing notes: D2.37

OC-2: Complete FAI Governance Record — Seventeen Record Categories

What the claim establishes. A complete FAI governance record comprising seventeen named record categories, covering the full lifecycle of an FAI event from participation authorization through dissolution and evolution feed routing. The seventeen categories span: participation authorization, aspect exchange scope, aspect contribution records, conflict records (preserved and resolved), orchestration rule logs, escalation records, dissolution records, evolution feed routing records, provenance chains, and post-event review records. A governance record that omits any category is incomplete in the prior-art sense: the omitted category represents an accountability gap that the architecture requires to be closed.

Prior-art significance. This claim establishes a specific, named completeness standard for FAI governance records. Seventeen categories of record are established as prior art. A record-keeping scheme that claims completeness for FAI governance must either meet these seventeen categories or demonstrate an architectural derivation from a different completeness criterion.

Establishing notes: D2.18

OC-3: Twenty-Test Operational Test Suite for FAI Compliance Verification

What the claim establishes. A twenty-test operational test suite that a governance record must pass for FAI compliance verification. The suite covers: substrate governance tests, participation authorization tests, aspect exchange tests, conflict-handling tests, dissolution tests, evolution feed tests, provenance tests, and retrospective auditability tests. The suite is designed so that passage of all twenty tests is sufficient to establish governance compliance for a given FAI event, and failure of any single test identifies a specific architectural gap.

Prior-art significance. This claim establishes a specific, enumerated compliance verification standard as prior art. A test suite claiming to verify FAI governance compliance is measured against these twenty tests. The structure of the suite — what it tests and why each test is included — is established prior art that an adversary cannot claim as novel without addressing D2.24.

Establishing notes: D2.24

OC-4: Twelve FAI Anti-Patterns — Named Governance Failure Modes

What the claim establishes. Twelve named FAI anti-patterns — specific governance failure modes that violate one or more of the Paper 3 architectural commitments. The twelve anti-patterns span: substrate governance failures (shared substrate operated without full Paper 1 commitments), participation failures (unauthorized or undocumented participation), aspect exchange failures (scope undefined or exceeded), conflict-handling failures (silent resolution, missing escalation, ungoverned preservation), dissolution failures (conditions unspecified or unmet), and evolution feed failures (routing undefined or ungoverned). Each anti-pattern is named, defined, and linked to the specific architectural commitment it violates.

Prior-art significance. Named anti-patterns are prior art for the failure modes they describe. A governance architecture that avoids or detects any of the twelve named failure modes cannot claim novelty for that avoidance or detection without addressing D2.19. The naming of failure modes also establishes that the modes are known — not novel discoveries waiting to be patented.

Establishing notes: D2.19

OC-5: FAI Onboarding and Offboarding Governance Sequences

What the claim establishes. Specific governance sequences for FAI onboarding (a Self entering participation in a shared substrate FAI event) and FAI offboarding (a Self exiting participation, with or without dissolution of the shared substrate). The onboarding sequence covers: participation authorization in home substrate, aspect exchange scope agreement, conflict-handling tier assignment, evolution feed routing specification, and governance record initialization. The offboarding sequence covers: dissolution condition verification, evolution feed hand-off confirmation, governance record closure, and home substrate update to reflect exit. Both sequences are derived from the Paper 3 architectural commitments and are fully specified prior art.

Prior-art significance. This claim establishes the onboarding and offboarding sequences as prior art. Any governance protocol for entering or exiting shared substrate FAI participation must address these sequences or demonstrate a derivation from a different set of architectural commitments.

Establishing notes: D2.69, D2.70

4. Category 3 — Governance Quality Claims (GC-1 through GC-5)

Governance quality claims state what properties the inter-Self coordination architecture guarantees about the character and quality of governance. These claims are architectural guarantees — they hold by design, not by procedural convention, and they are inherited from the Paper 1 and Paper 2 commitments the architecture builds on. Each governance quality claim names a specific property, states what guarantees it, and identifies the establishing notes that derive it at inter-Self scope.

GC-1: Non-Specialist Governance Applies at FAI Scope

What the claim establishes. Operational practitioners — not specialists in AI architecture, inter-organizational coordination theory, or substrate engineering — can govern a compliant FAI event using the governance records and orchestration rules the architecture requires. Non-specialist governance is not a concession or a simplification; it is a design property of the architecture. The shared substrate carries the coordination knowledge and governance semantics that make expert judgment exercisable by non-specialists following the orchestration rules. This property holds at inter-Self scope by the same derivation that establishes non-specialist governance within a single Self.

Prior-art significance. This claim blocks any attempt to patent non-specialist-accessible FAI governance as a novel usability feature. Non-specialist governance at FAI scope is derived from the Paper 1 and Paper 2 commitments; it is prior art.

Establishing notes: D2.75

GC-2: Determinism Contract Holds at FAI Scope

What the claim establishes. FAI governance decisions are deterministic in the following sense: given the governance record for an FAI event and the orchestration rules that applied during that event, any practitioner can reproduce the governance decisions that were made — without requiring access to participants not present at the original event, and without requiring interpretation of undocumented judgment calls. The determinism contract holds at inter-Self scope because the governance record requirements (OC-2) and the orchestration rule logging requirements guarantee that all decision-relevant information is captured in the record. The contract applies to conflict-handling decisions, aspect exchange decisions, dissolution decisions, and evolution feed routing decisions.

Prior-art significance. This claim establishes the determinism contract as a named architectural property at inter-Self scope. A governance architecture claiming auditability or reproducibility for inter-organizational coordination must contend with this prior-art formulation of what determinism requires and how the architecture achieves it.

Establishing notes: D2.66

GC-3: Path Retracement at Inter-Self Scope — Four-Link Authorization Chain

What the claim establishes. Every FAI governance decision is retraceable through a four-link cross-organizational authorization chain: (1) the specific governance action taken; (2) the orchestration rule that authorized it; (3) the substrate content that specified that rule; (4) the human authority that authored or approved that substrate content. All four links are present in a complete governance record. Path retraceability at inter-Self scope means that the authorization chain crosses organizational boundaries — the chain includes authorities in the home substrates of each participating Self — without breaking the four-link structure.

Prior-art significance. This claim establishes path retraceability with a specific four-link structure as prior art for inter-organizational coordination architectures. A system claiming path retraceability or authorization-chain auditability for inter-organizational AI coordination must address this prior-art formulation.

Establishing notes: D2.03, D2.67

GC-4: Home Perimeter Integrity During FAI

What the claim establishes. FAI participation does not alter the home governance architecture of any participating Self. A Self's home substrate governance commitments — the Paper 1 and Paper 2 commitments that define its internal coordination architecture — remain fully intact throughout and after FAI participation. The shared substrate is a temporary construction at inter-Self scope; its existence and operation do not modify, override, or supersede the home governance commitments of any participant. Home perimeter integrity is an architectural guarantee, not a procedural promise: the design of the architecture makes home-perimeter modification via FAI impossible, not merely unlikely.

Prior-art significance. This claim blocks any attempt to patent home perimeter preservation during inter-organizational AI coordination as a novel safety or isolation feature. Home perimeter integrity follows from the architecture's treatment of the shared substrate as a temporary, bounded construction operating at a scope above the home substrates — not as an overlay or replacement of them.

Establishing notes: D2.17

GC-5: Governance Sovereignty Protections — Exit, Withdrawal, and Non-Delegation

What the claim establishes. Three named governance sovereignty protections that the architecture guarantees each participating Self: (1) exit rights — a Self may exit FAI participation at any time, subject only to dissolution conditions it consented to in its home substrate; (2) withdrawal rights — a Self may withdraw aspects or reduce participation scope without triggering full dissolution, provided withdrawal conditions are specified in the shared substrate; (3) the non-delegation principle — a Self's home governance authority cannot be delegated to the shared substrate or to other participants; decisions requiring home authority must be routed back to the home substrate for resolution. All three protections are substrate-content configurable but not substrate-content eliminable: they may be parameterized but not removed.

Prior-art significance. This claim establishes specific, named sovereignty protections as prior art for inter-organizational AI coordination architectures. A coordination framework claiming to preserve participant sovereignty or governance autonomy must address these three protections and their architectural derivation.

Establishing notes: D2.27, D2.43, D2.46

5. Category 4 — Population-Scope Claims (PC-1 through PC-3)

Population-scope claims address what the architecture commits to at the scale of many Selves participating in FAI across organizational boundaries over time. These claims are stated in the calibrated-humility register: they are architectural in character — derived from the Paper 3 commitments — but population-empirical in verification. The prior-art value of calibrated-humility claims is that they establish the architecture's position on population-scale phenomena, blocking claims of novelty for that position, without overreaching into empirical assertions the corpus cannot establish.

PC-1: FAI Network Topology is a Distributed Governance Phenomenon

What the claim establishes. The topology of a network of Selves engaged in FAI — which Selves participate with which, at what aspect exchange scope, and under what governance terms — is determined by the distributed governance decisions of the participating Selves, not by any central authority. No central coordinator determines the topology. Each bilateral or multilateral FAI event is authorized in the home substrates of the participating Selves; the network topology is the aggregate of those distributed authorizations. This is not an empirical observation about any particular deployment; it is an architectural consequence of the design commitment that FAI configuration is substrate content in each participating Self's home substrate.

Prior-art significance. This claim blocks any attempt to patent distributed FAI topology determination as a novel network architecture. The distributedness of topology determination is a direct consequence of the AC-5 and GC-5 prior art.

Establishing notes: D2.61

PC-2: Pattern Propagation is Governed at Each Step

What the claim establishes. When a coordination pattern propagates across a population of Selves via FAI — when a pattern that originated in one Self’s home substrate is exchanged via FAI, enters another Self’s evolution feed, and influences that Self’s evolved state — the propagation is governed at each step. The exchange step is governed by the FAI governance sequence. The ingestion step is governed by the receiving Self’s home evolution machinery under its home orchestration rules. The evolution step is governed by the receiving Self’s home governance commitments. Collective evolution via FAI is not ungoverned emergence; it is a chain of governed steps, each of which leaves a governance record.

Prior-art significance. This claim blocks any attempt to characterize FAI-mediated collective evolution as emergent or ungoverned and then patent a governance overlay for it. The governance at each step is inherent in the architecture, not added on top of an ungoverned propagation mechanism.

Establishing notes: D2.32, D2.62

PC-3: Each Self’s Home Governance Authority is Preserved at Population Scope

What the claim establishes. At population scope — across many Selves, many FAI events, and extended time periods — each Self’s home governance authority remains intact and substrate-content configurable. No population-scale coordination dynamic removes, overrides, or erodes a Self’s authority over its home substrate. Population-scale coordination is the aggregate of individual FAI events, each of which satisfies the GC-5 sovereignty protections. The calibrated-humility position of this claim is that the architecture guarantees preservation by design; population-scale empirical patterns cannot override architectural design commitments.

Prior-art significance. This claim blocks any attempt to patent governance authority preservation at population scale as a novel feature of a multi-Self coordination system. Preservation of home governance authority is a direct consequence of AC-1 (shared substrate does not supersede home substrates), GC-4 (home perimeter integrity), and GC-5 (non-delegation principle).

Establishing notes: D1.26, D2.31

6. Using This Index

Locating a claim. Each claim is identified by its category prefix and number (AC-1 through PC-3). The establishing note references are listed at the end of each claim entry. Establishing notes are identified by their D-series IDs; each D-series note is a standalone document in the Phase D2 corpus and can be located by its ID.

Reading the establishing notes. Each establishing note provides the formal derivation, operational detail, and prose argument for its claim. This index does not summarize that content; it indexes it. For a complete understanding of any claim, read the establishing notes cited for that claim.

Adversarial use. An adversary challenging any Phase D2 prior-art claim must address: (a) the claim as stated in this index; (b) the establishing notes cited; and (c) the Paper 1, Paper 2, and Paper 3 commitments from which the claim is derived. The fourteen claims are not independent — the architectural claims (Category 1) underpin the operational claims (Category 2), which in turn support the governance quality claims (Category 3); the population-scope claims (Category 4) follow from the architectural and governance quality claims. A challenge that addresses one claim must account for its dependencies on the others.

What follows. Note D2.80 is the final Phase D2 note. It provides full synthesis of the Phase D2 corpus — a unified account of what the seventy-eight operational notes and this navigation index collectively establish, how the fourteen claims relate to each other and to the Paper 3 source commitments, and a formal closure statement for the phase. Phase D3 (anti-pattern formalizations) and Phase D4 (composition pairs) proceed from the prior-art foundation established by Phases D0, D1, and D2.

Note D2.79 — #574 in the CKS derivation series. Phase D2 complete with D2.80 to follow.